

Erlang programs.

The following is the code of a simple Erlang module, called *circle* and is saved in a file called *circle.erl* that calculates the circumference and the area of a circle, given its radius:

```
%%% Filename: circle.erl
%%% Programmer: Mihalis Tsoukalos
%%% Date: Saturday 25 May 2013
-module(circle).
-export([length/1, area/1]).
length(Radius) -> 2 * 3.1415 * Radius.
area(Radius) -> 3.1415 * 3.1415 * Radius.
```

At the top of the module, the *-module* directive gives the name of the module, which is *circle*. The *-export* directive defines the list of functions that are visible to the world. The *circle* module exports two functions called *length* and *area*. Note that both the name and the total number of parameters of a function are defined.

Erlang considers functions with the same name but different number of parameters as different functions. All module code must be included in functions.

In order to compile the *circle.erl* file, you should execute the following command from the Erlang shell:

```
5> c(circle).
{ok,circle}
```

If you now look inside the directory where *circle.erl* resides, you will also see a *binary* file called *circle.beam*, which is the compiled module code in BEAM format. Listing the available functions as well as other useful information about a module can be done using the following command:

```
7> circle:module_info().
[{exports,[[length,1],
             {area,1},
             {module_info,0},
             {module_info,1}]],
 {imports,[]},
 {attributes,[{vsn,[187948631488141693241015239821467106334]}],
 {compile,[{options,[]},
            {version,"4.9.1"},
            {time,[2013,5,25,14,16,26]},
            {source,"/Users/mtsouk/docs/article/Erlang.OSFY/
circle.erl"}]}}]
```

Using its functions can be done as follows:

```
9> circle:area(1).
9.86902225
10> circle:length(1).
```

```
6.283
11> circle:length(2).
12.566
12>
```

If you use *radius* instead of *Radius* as the variable name, you will get the following warning message while compiling the *circle* module:

```
3> c(circle).
circle.erl:8: Warning: this expression will fail with a
'badarith' exception
circle.erl:10: Warning: this expression will fail with a
'badarith' exception
{ok,circle}
```

This happens because, as mentioned earlier, *variables* start with capital letters in Erlang.

Last, if you want to use the *area* function from the *circle* module in another module that you create, you should include the following command inside the new module:

```
-import(circle, [area/1]).
```

There is no command to import all functions at once from an existing Erlang module.

Documenting Erlang code

Erlang allows you to include documentation during the module writing. You should always document your code not only to help other programmers who may read it, but even for yourself, because you may need to change or debug your code after some time. The full and documented version of the *circle* module, which is now called *circleDoc*, is as follows:

```
%@author Mihalis Tsoukalos <tsoukalos@sch.gr> [http://
users.sch.gr/tsoukalos]
%@doc This is a module that provides functions for
calculating the length
%and the area of a circle.
%@reference For an article for <a href="http://www.
linuxforu.com">Open Source for You</a>.
%©copyright 2013, Mihalis Tsoukalos
%@version 1.0
-module(circleDoc).
-export([length/1, area/1]).
%@doc Calculates the length of a circle given its radius
%using the 2 * pi * R mathematics formula.
length(Radius) -> 2.0 * 3.1415 * Radius.
%@doc Calculates the area of a circle given its radius
%using the pi * pi * R mathematics formula.
area(Radius) -> 3.1415 * 3.1415 * Radius.
```

In order to create the HTML files, you should execute the